

NM4QM 2026 · HANDS-ON SESSION · 45 MIN

Spin and anomalous Hall effects with WannierBerri and GPAW

Symmetry-adapted Wannier functions, spin–orbit coupling on demand, the spin Hall effect of graphene and the anomalous Hall effect of the altermagnet MnTe

Yaroslav Zhumagulov

Chair of Computational Physics of Condensed Matter (C3MP), EPFL, Lausanne, Switzerland

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Two puzzles, two acts

0–8 min

Setup and ideas

8–19

Act 1: graphene

19–37

Act 2: MnTe

37–45

Your material,
questions

TWO PUZZLES

- Spin–orbit coupling turns graphene into a quantum spin Hall insulator with a gap of $42 \mu\text{eV}$
- MnTe has no net magnetisation, yet shows an anomalous Hall effect

WHAT YOU WILL DO

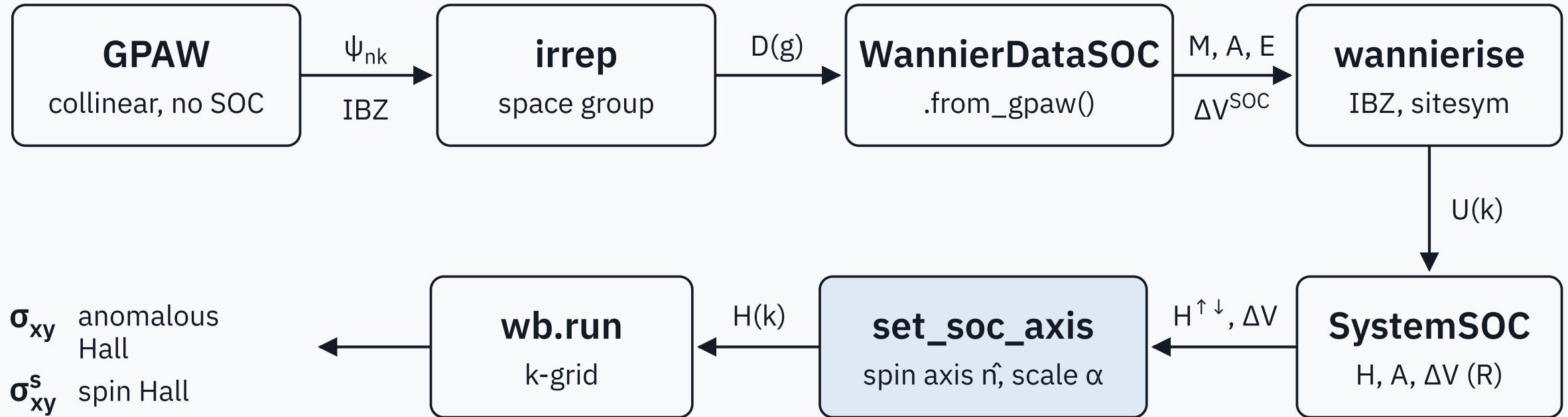
- Build symmetry-adapted Wannier functions straight from GPAW
- Switch spin–orbit coupling on, off and around, without new DFT
- Compute spin and anomalous Hall conductivities

Before we start: one check, 20 seconds

```
cd wannier-berri-demo && conda activate wbdemo
python setup/check_setup.py           # should end with "Ready."
cd 01_graphene
mpirun -np 4 python 1_dft.py          # 10 s
jupyter lab graphene.ipynb
```

- No environment yet: README, "Before the session"
- DFT fails: unpack our `dft-files.tar.gz` and carry on
- Fell behind: `bash run_all.sh` (6 min); finished notebooks in `reference/`
- Numbers on the slides are checkpoints: compare them with yours

One Python pipeline, GPAW to conductivities



Each box is one call in the notebooks. The shaded step needs no new DFT and no new Wannierisation.

ψ_{nk} : Bloch states · $D(g)$: symmetry matrices on them · M, A, E: overlaps, projections, energies · ΔV^{SOC} : SOC matrix elements from the PAW spheres · $U(k)$: Wannier gauge · $H, A, \Delta V(\mathbf{R})$: Hamiltonian, Berry connection and SOC in the Wannier basis · $\hat{n}, \alpha$: spin axis, SOC scale · σ_{xy} : anomalous Hall (charge current), σ_{xy}^s : spin Hall (spin current) conductivity

Berry curvature needs millions of k-points; DFT gives tens

Intrinsic anomalous Hall conductivity:

$$\sigma_{xy} = -\frac{e^2}{\hbar} \int_{\text{BZ}} \frac{d^3k}{(2\pi)^3} \sum_n f(\varepsilon_{n\mathbf{k}}) \Omega_n^z(\mathbf{k})$$

$$\Omega_n^z = -2 \text{Im} \sum_{m \neq n} \frac{v_{nm}^x v_{mn}^y}{(\varepsilon_n - \varepsilon_m)^2}$$

f : occupation, ε_n : band energy, $v_{nm}^a = \langle u_n | \partial_{k_a} H | u_m \rangle$. Ω_n^z is large and narrow wherever two bands come close.

Wannier functions give $H(\mathbf{k})$ at any $\mathbf{k}$:

$$H_{ij}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} \langle w_{i0} | \hat{H} | w_{j\mathbf{R}} \rangle$$

$w_{j\mathbf{R}}$: Wannier function j in the cell at $\mathbf{R}$; a few hundred $\mathbf{R}$ are enough.

7 → 1.44 M

graphene: DFT k-points → spin
Hall grid

24 → 0.59 M

MnTe: DFT k-points →
anomalous Hall grid

30 s, 1 min

the two integrals on an 8-core
laptop

Symmetry-adapted Wannier functions: DFT on irreducible k only

Each Wannier function should transform like its orbital:

$$\hat{g} w_{i\mathbf{R}}(\mathbf{r}) = \sum_j D_{ji}(g) w_{j,g\mathbf{R}}(\mathbf{r})$$

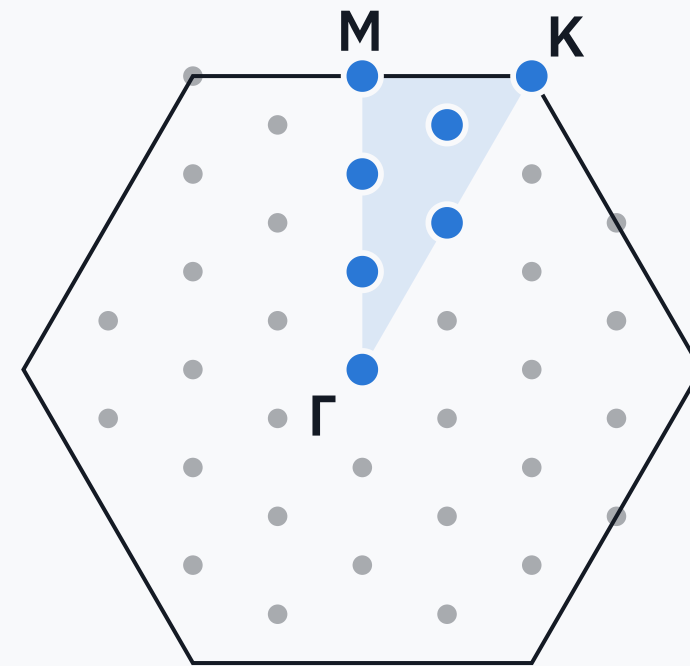
$\hat{g}$: symmetry operation, $D(g)$: its matrix on the orbitals of one site

Then the gauge at $g\mathbf{k}$ follows from $\mathbf{k}$ (Sakuma 2013):

$$U(g\mathbf{k}) = \tilde{d}(g, \mathbf{k}) U(\mathbf{k}) D(g, \mathbf{k})^\dagger$$

U : Bloch $\rightarrow$ Wannier gauge, $\tilde{d}$: g acting on the Bloch states (from irrep)

- DFT on 7 of 36 k-points for graphene, 24 of 144 for MnTe
- Degeneracies come out exact



6 × 6 grid: 36 points, 7 irreducible (blue)

Wannierisation: WannierBerri vs Wannier90

$$\Omega = \sum_n \left[\langle w_n | r^2 | w_n \rangle - \left| \langle w_n | \mathbf{r} | w_n \rangle \right|^2 \right] = \Omega_I + \tilde{\Omega}$$

Ω : spread of the Wannier functions · Ω_I : fixed by the chosen subspace (disentanglement) · $\tilde{\Omega}$: depends on the gauge $U(\mathbf{k})$ (localisation)

	Wannier90	WannierBerri, this demo
Input	.amn, .mmn, .eig files from a DFT interface	computed in Python from the GPAW wave functions
k-points	full grid; irreducible ones with site_symmetry	irreducible ones
Algorithm	disentangle first, then minimise $\tilde{\Omega}$ by steepest descent or conjugate gradients	one loop: a disentanglement step and a local gauge update per iteration, no gradient descent
Symmetry	site_symmetry with a .dmn file from pw2wannier90, not with a frozen window	from irrep, for any code it reads; frozen window allowed
SOC	spinor DFT run with SOC	collinear run, SOC added afterwards; spin ↓ from symmetry in altermagnets

Post-processing: WannierBerri vs postw90

$$H(\mathbf{K} + \boldsymbol{\kappa}) = \sum_{\mathbf{R}} e^{i\boldsymbol{\kappa} \cdot \mathbf{R}} \left[e^{i\mathbf{K} \cdot \mathbf{R}} H(\mathbf{R}) \right]$$

$\boldsymbol{\kappa}$: points of a coarse FFT grid · $\mathbf{K}$: shifts of that grid; one FFT per shift, the shifts together give the dense grid

	postw90	WannierBerri
$H(\mathbf{R}) \rightarrow H(\mathbf{k})$	sum over $\mathbf{R}$ at every $\mathbf{k}$, cost $\propto N_{\mathbf{R}}$	FFT over sub-grids, cost $\propto \log N_{\text{FFT}}$ per $\mathbf{k}$
Symmetry	full grid	irreducible points of the magnetic point group, result symmetrised
Hot spots	finer mesh where $ \Omega $ exceeds a threshold	recursive adaptive refinement
Fermi levels	cost grows with their number	~1000 at almost no extra cost
SOC axis	fixed by the DFT run	<code>set_soc_axis</code> : any axis and strength

Up to three orders of magnitude faster on dense grids. In this demo: 1401 Fermi levels in one run; for $\hat{\mathbf{n}} \parallel \mathbf{y}$ only the $\mathbf{k}$ -points irreducible under $m'm'm$ (8 operations) are computed.

Does graphene have a gap?

Step	Run	Checkpoint
0 · DFT	<code>mpirun -np 4 python 1_dft.py</code>	10 s, <code>gpaw/nscf.gpw</code>
1 · Symmetry	<code>irrep</code>	48 operations, 7 of 36 k-points
2 · Wannier functions	<code>wannierise</code>	spread 0.91 Å ² each
3 · Bands	<code>interpolation</code>	blue on grey in the window
4 · SOC on demand	<code>set_soc_axis</code>	gap 35.8 μeV (GPAW 38.7)
5 · Spin Hall	<code>wb.run</code> , 1200 ² k	1.0003 e/2π, 30 s

Graphene: DFT in 10 s, two p_z Wannier functions

- PBE, 400 eV, $6 \times 6 \times 1$ grid, 15 Å vacuum, no SOC
- NSCF only on the 7 irreducible k-points
- Frozen window $E_F - 2.9 \dots E_F + 2.8$ eV, plus the π state at Γ

0.91 Å²

spread per function

48

symmetry operations

3 s + 2 s

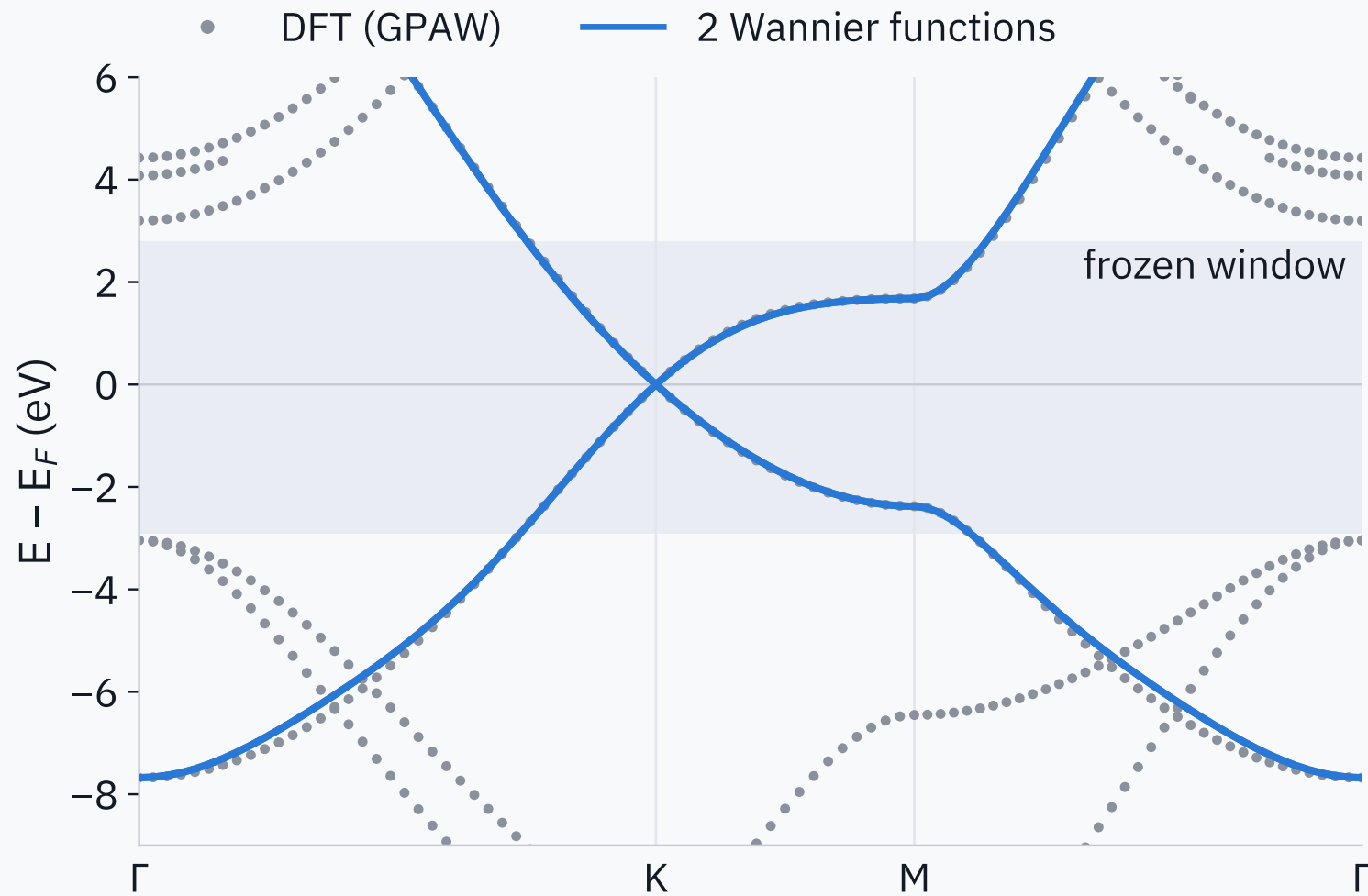
from_gpaw + wannierise

```
calc = GPAW(mode=PW(400), xc="PBE",
             kpts={"size": (6, 6, 1), "gamma": True})
atoms.calc = calc
atoms.get_potential_energy()
sg = SpaceGroup.from_gpaw(calc) # irrep
kpts = sg.get_irreducible_kpoints_grid((6, 6, 1)) # 7 of 36
calc.fixed_density(kpts=kpts, nbands=20).write(
    "gpaw/nscf.gpw", mode="all")

# graphene.ipynb
pz = Projection(position_num=[[1/3, 2/3, 0], [2/3, 1/3, 0]],
                orbital="pz", spacegroup=sg)
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=ProjectionsSet([pz]))
wandata.wannierise(froz_min=EF - 2.9, froz_max=EF + 2.8,
                  frozen_states={0: [1]}, # pi at Gamma
                  num_iter=100, sitesym=True)
system = wb.SystemSOC.from_wannierdata(wandata, berry=True)
```

Two Wannier bands follow DFT through the frozen window

- Grey: GPAW along Γ –K–M– Γ , not used in the fit
- Blue: the two-band Wannier model
- Shaded: frozen window



SOC as a matrix you can rotate and scale

$$H(\mathbf{k}) = H_0(\mathbf{k}) \otimes \sigma_0 + \alpha \sum_c \Delta V^c(\mathbf{k}) \otimes \sigma_c(\hat{\mathbf{n}})$$

$$\Delta V_{mn}^c(\mathbf{k}) = \sum_{a,ij} \langle \tilde{\psi}_{m\mathbf{k}} | \tilde{p}_i^a \rangle \langle \phi_i^a | \xi_a(r) L_c | \phi_j^a \rangle \langle \tilde{p}_j^a | \tilde{\psi}_{n\mathbf{k}} \rangle$$

$$\xi_a(r) = \frac{1}{4m^2c^2} \frac{1}{r} \frac{dV_a}{dr}$$

H_0 : Wannier Hamiltonian without SOC · $\sigma_c(\hat{\mathbf{n}})$: Pauli matrices, z along the spin axis $\hat{\mathbf{n}}$ · α : SOC scale, 1 = real · $\tilde{\psi}, \tilde{p}_i^a, \phi_i^a$: PAW pseudo wave functions, projectors, partial waves of atom a · L_c : orbital angular momentum · V_a : potential in sphere a

- Same PAW matrix elements as GPAW's own SOC
- New $\hat{\mathbf{n}}$ or α : no new DFT

```
K = [1/3, 1/3, 0]
energy = wb.calculators.tabulate.Energy(
    degen_thresh=1e-8)
for alpha in [0, 1]:
    system.set_soc_axis(alpha_soc=alpha)
    E = wb.evaluate_k(system, k=K,
        calculators={"E": energy})
```

	E(K) - E_F (ueV)				gap
alpha = 0	28.6	28.6	28.6	28.6	0.0
alpha = 1	10.7	10.7	46.5	46.5	35.8
GPAW SOC	9.0	9.0	47.7	47.7	38.7

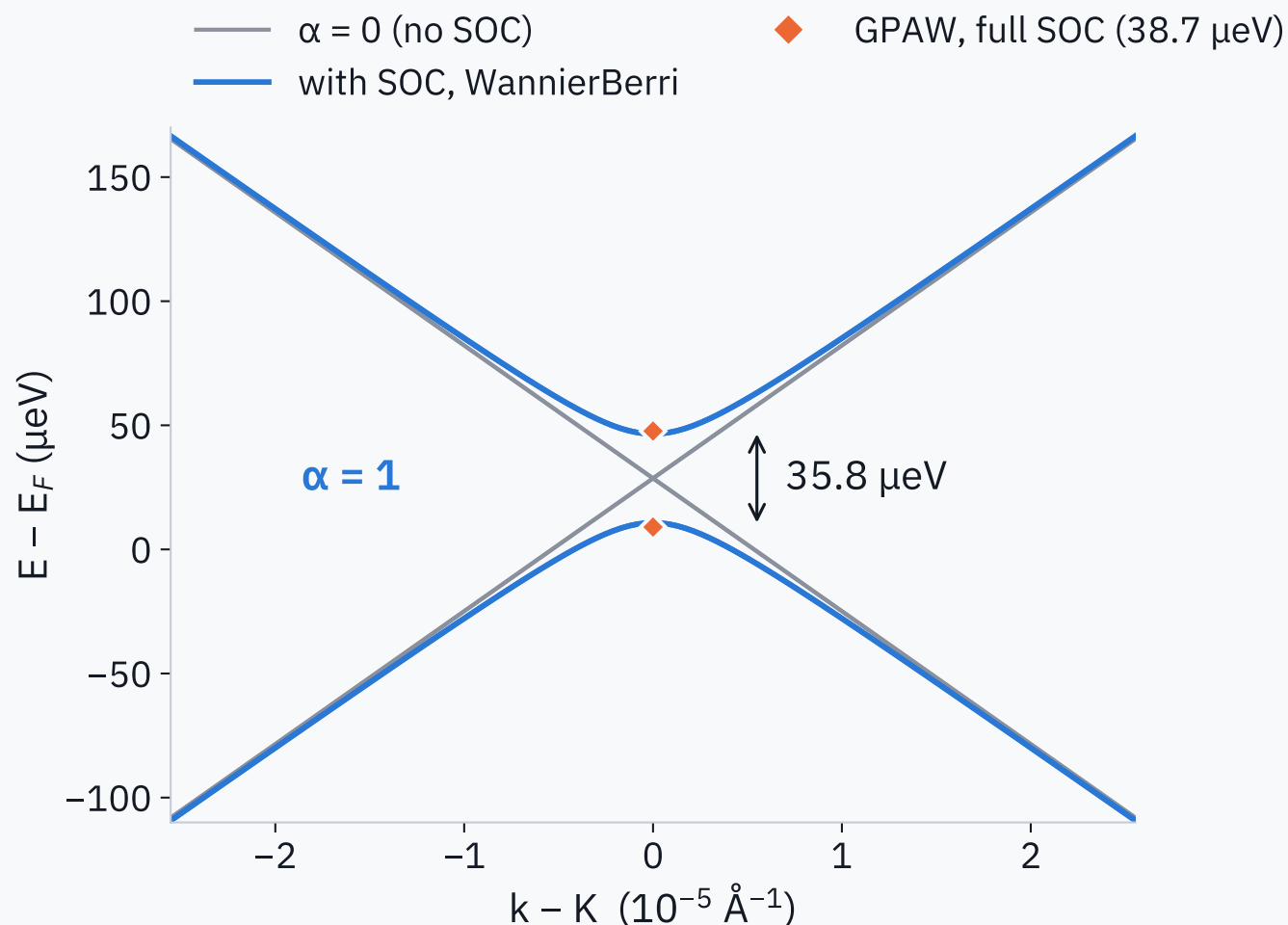
`degens_thresh=1e-8`: the default 10^{-4} eV would merge the split pair

Kane–Mele gap: 36 μeV from two Wannier functions

Gap at K: **35.8 μeV**

GPAW, full SOC, 20 bands: **38.7 μeV**

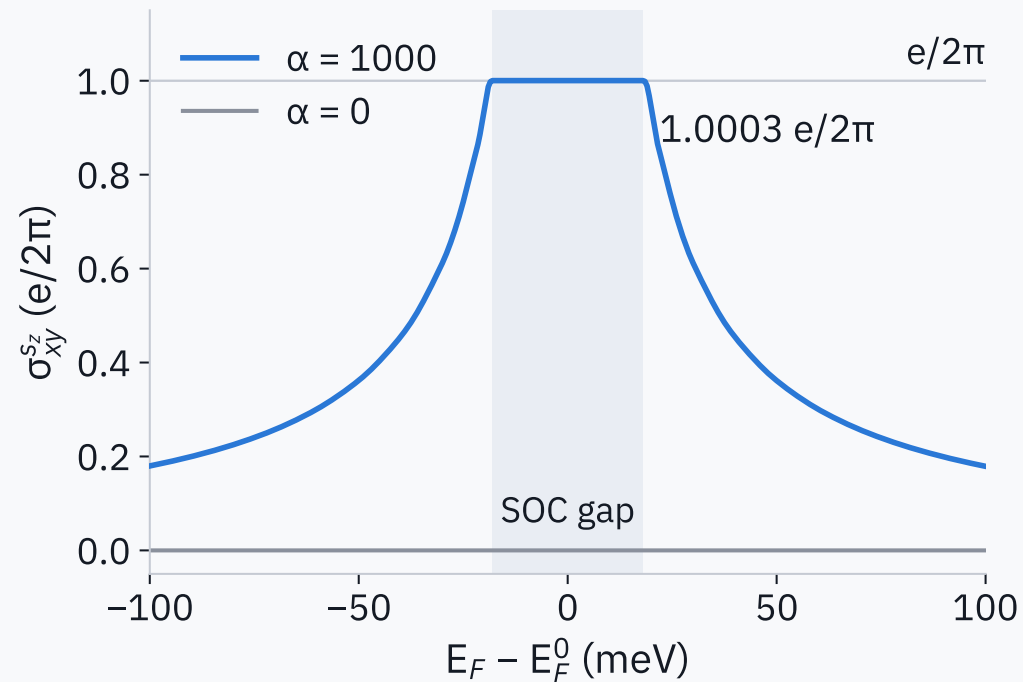
- p_z has $m = 0$, so the gap comes from the small d part of p_z in the PAW spheres (Gmitra 2009)
- Missing 3 μeV : second-order SOC via the σ bands
- Depends on cutoff and PAW dataset; experiment 42.2 μeV (Sichau 2019)



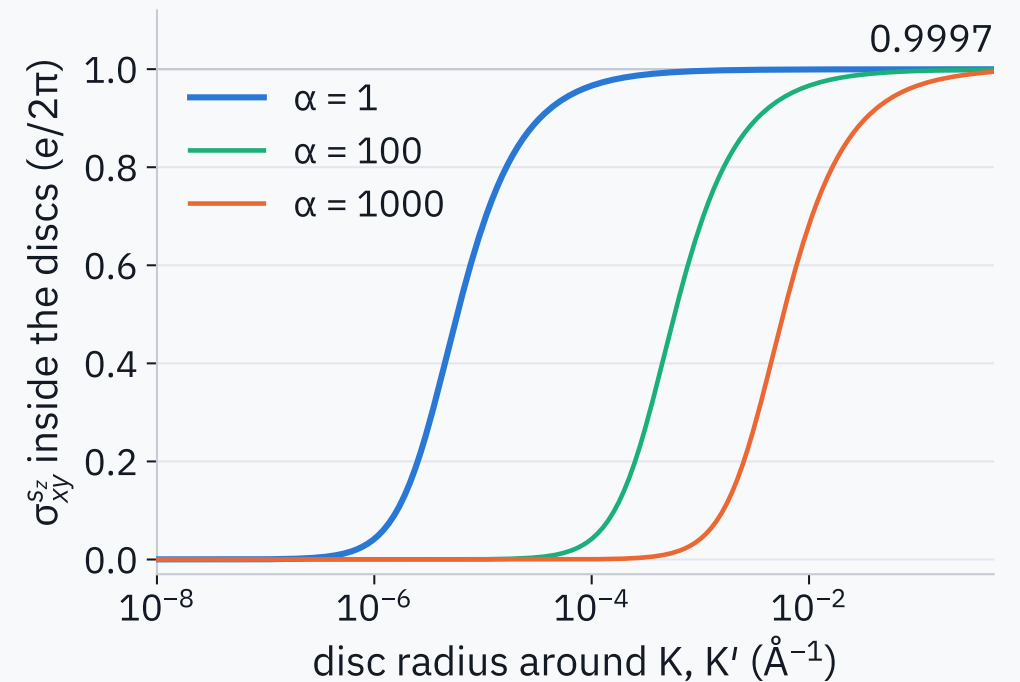
Spin Hall conductivity: $e/2\pi$, also at the real gap

$$\sigma_{xy}^{sz} = \frac{e}{\hbar} \int \frac{d^2k}{(2\pi)^2} \sum_n f_{n\mathbf{k}} \Omega_n^{sz}, \quad \Omega_n^{sz} = -2\hbar^2 \operatorname{Im} \sum_{m \neq n} \frac{(j_x^{sz})_{nm} (v_y)_{mn}}{(\varepsilon_n - \varepsilon_m)^2}$$

$j_x^{sz} = \frac{1}{2} \{s_z, v_x\}$: spin current, v : velocity. With s_z conserved, $\sigma_{xy}^{sz} = (e/2\pi)(C_\uparrow - C_\downarrow)/2 = e/2\pi$



$\alpha = 1000$, 1200^2 grid with tetrahedra: $1.0003 e/2\pi$



$\alpha = 1$ (36 μeV), log-polar grid around K and K': $0.9997 e/2\pi$

A Hall effect without magnetisation

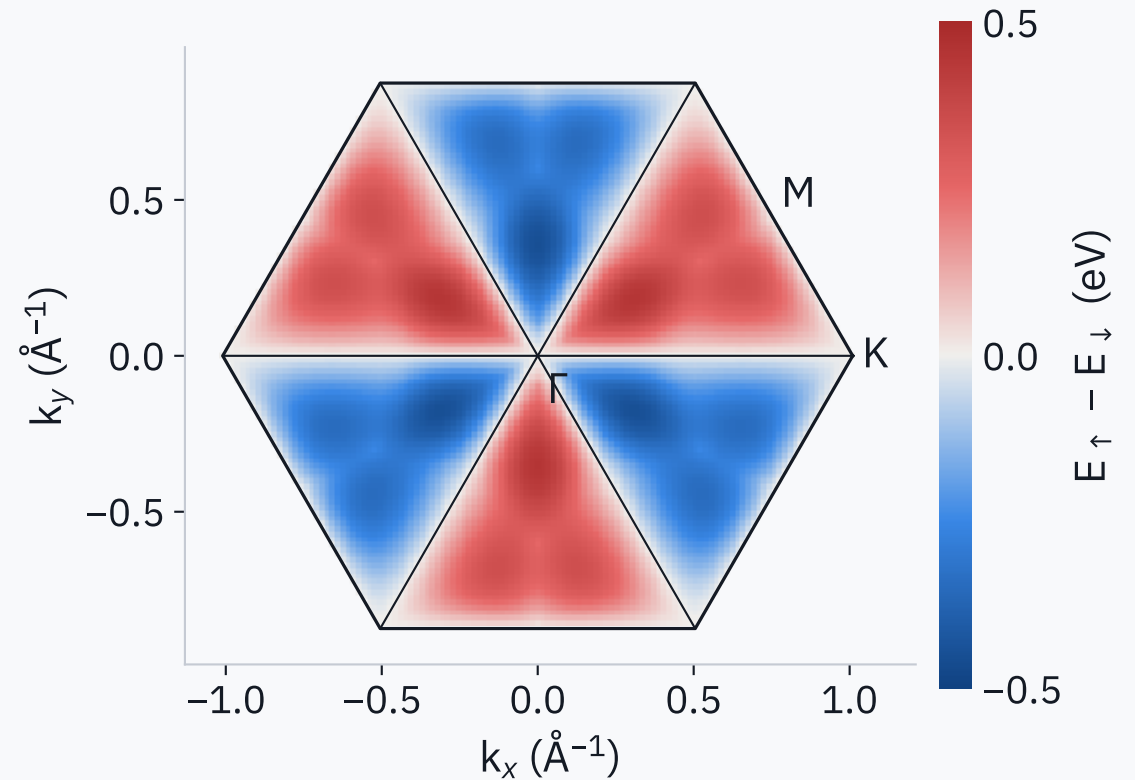
Step	Run	Checkpoint
0 · DFT	<code>mpirun -np 4 python 1_dft.py</code>	45 s, moments $\pm 4.41 \mu_B$
1–2 · Spin $\uparrow$ Wannier	<code>altermagnetic=True</code>	24 of 144 k-points, 25 s
3 · Splitting	no SOC	up to 0.97 eV at $k_z \neq 0$
4 · Turn $\hat{n}$	<code>set_soc_axis</code>	shift at A: 322 $\rightarrow$ 75 meV
5 · Anomalous Hall	three $\hat{n}$	0 for x and c, up to 194 S/cm for y
6 · Rotate in plane	0–30°	$\sigma(5^\circ)/\sigma(30^\circ) = 0.56$

Altermagnets: zero net moment, spin-split bands without SOC

- Two sublattices related by a rotation or mirror, not by a translation or inversion
- Bands spin split without SOC; the sign alternates in $\mathbf{k}$ (d-, g-, i-wave)
- No PT or tT symmetry, so with SOC an anomalous Hall effect is possible, depending on the Néel vector
- **MnTe**: NiAs type, $T_N \approx 310$ K, gap 1.27 eV

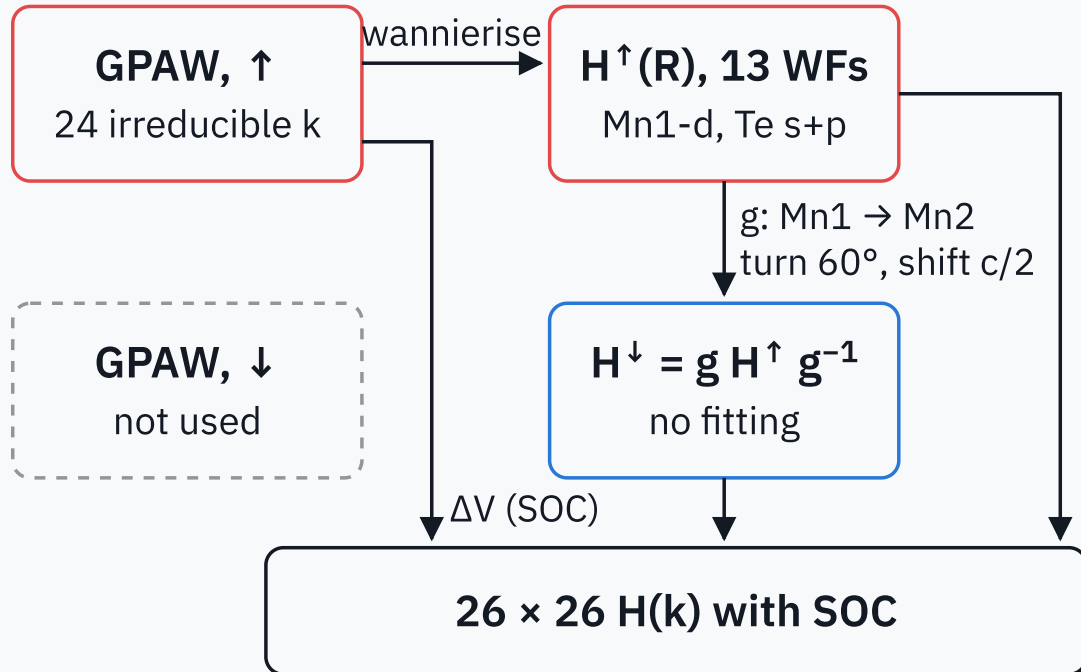
$$[C_2 \parallel g] : E_{\uparrow}(\mathbf{k}) = E_{\downarrow}(R_g \mathbf{k})$$

C_2 : flips all spins, g : swaps the Mn sublattices, R_g : its rotation part



MnTe, no SOC: $E_{\uparrow} - E_{\downarrow}$ at $k_z = 0.35 \text{ \AA}^{-1}$

Wannierise spin $\uparrow$ only, spin $\downarrow$ follows by symmetry



PBE+U ($U = 4$ eV), $6 \times 6 \times 4$, 45 s: moments $\pm 4.41 \mu_B$, gap 0.77 eV (exp. 1.27–1.41 eV; a larger U does not fix it)

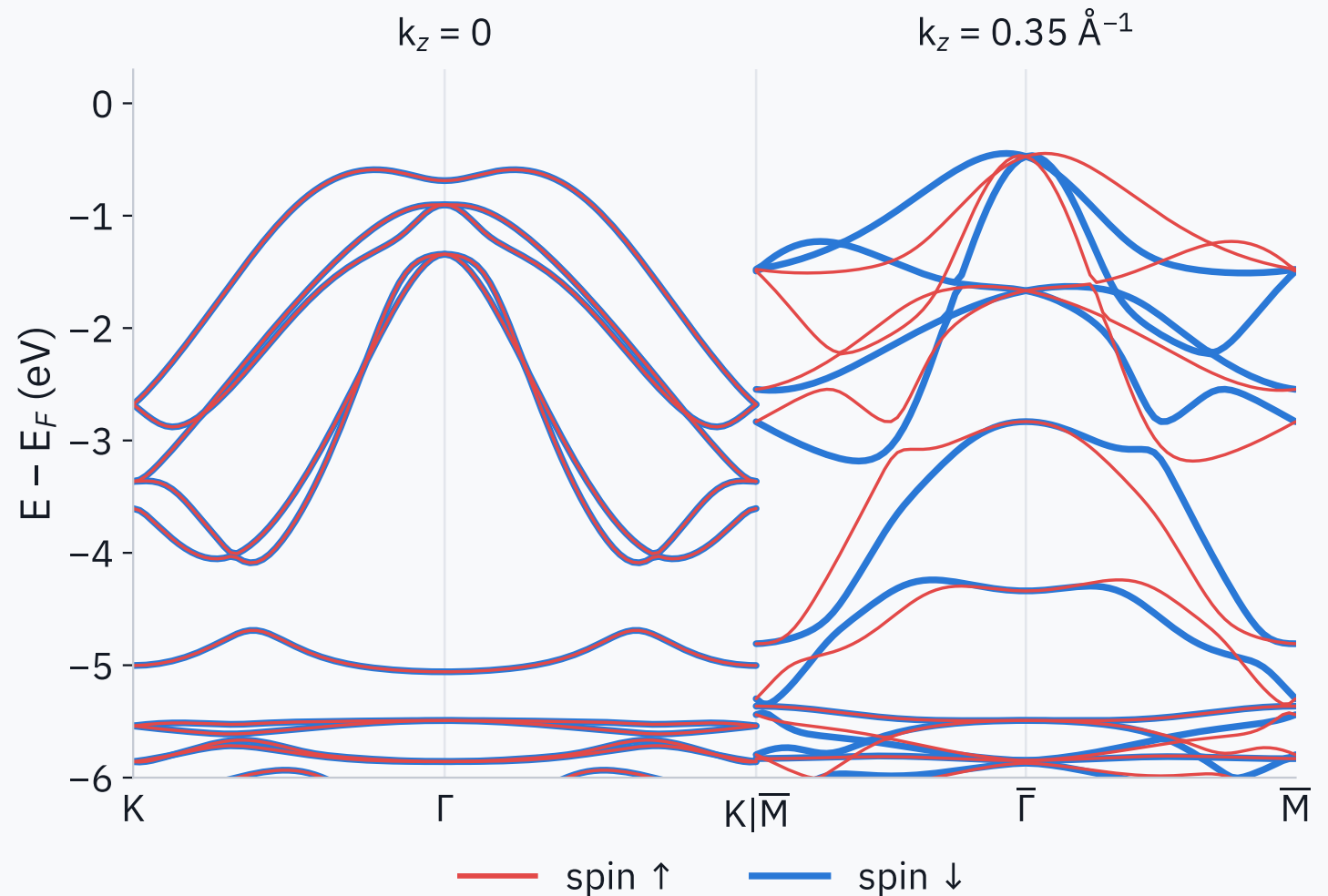
```

calc = GPAW("gpaw/nscf.gpw") # PBE+U, 24 of 144 k
sg = SpaceGroup.from_gpaw(calc)
Te = [[1/3, 2/3, 1/4], [2/3, 1/3, 3/4]]
projections = ProjectionsSet([
    Projection(position_num=[0, 0, 0], orbital="d",
               spacegroup=sg), # Mn1 only
    Projection(position_num=Te, orbital="sp2",
               spacegroup=sg, xaxis=[0, -1, 0],
               rotate_basis=True),
    Projection(position_num=Te, orbital="pz",
               spacegroup=sg)])

wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=projections,
    altermagnetic=True, # spin down from symmetry
    IBstart=8) # skip Mn 3s, 3p
wandata.wannierise(froz_min=-np.inf,
                  froz_max=EF + 0.1,
                  num_iter=200, sitesym=True)
system = wb.SystemSOC.from_wannierdata(wandata,
                                       berry=True)
  
```

No SOC: degenerate at $k_z = 0$, split by up to 1 eV at $k_z \neq 0$

- Red: spin $\uparrow$, Wannierised
- Blue: spin $\downarrow$, from symmetry
- At $k_z = 0.35 \text{ \AA}^{-1}$ up to 0.97 eV apart, with opposite order on the two sides of Γ : g-wave



With SOC the Néel vector picks the magnetic point group

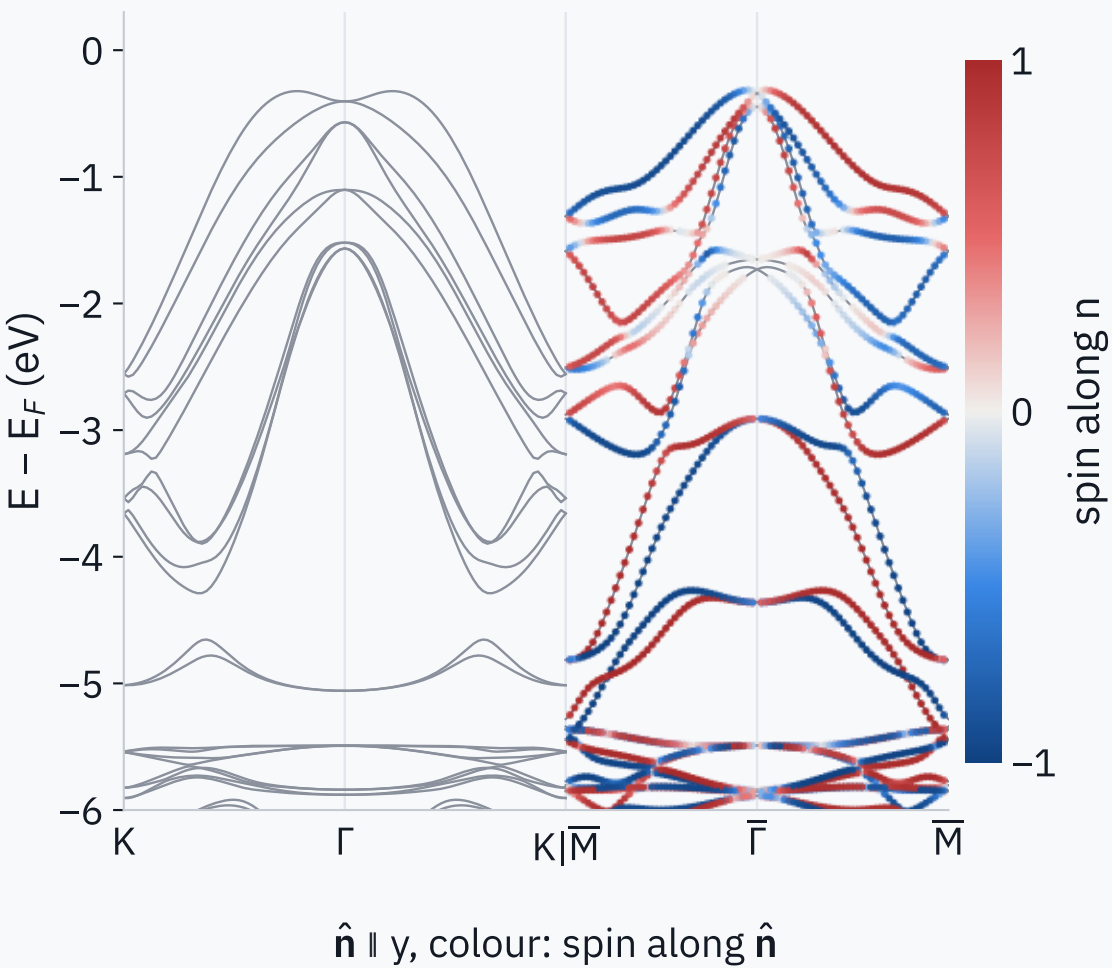
Néel vector $\hat{n}$	Group	Hall vector
$\parallel c$	$6'/m'm'm$	none
$\parallel a = x$	mmm	none
$\perp a = y$	$m'm'm$	$\sigma_{xy} \neq 0$

$$g : \sigma_H \rightarrow \det(R_g) R_g \sigma_H$$

$$\mathcal{T} : \sigma_H \rightarrow -\sigma_H$$

$\sigma_H = (\sigma_{yz}, \sigma_{zx}, \sigma_{xy})$: Hall vector · R_g : rotation part of g · $\mathcal{T}$: time reversal · primed: combined with $\mathcal{T}$

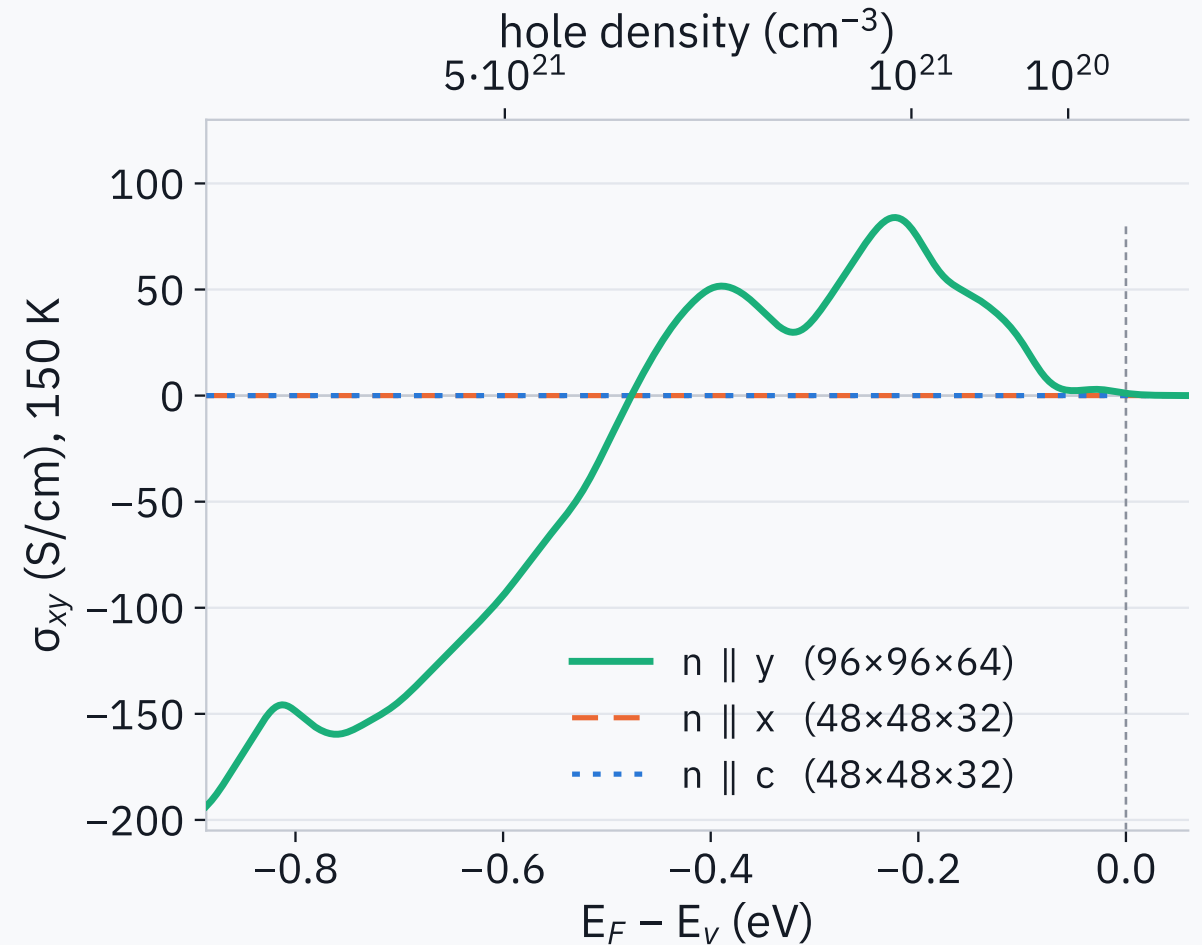
SOC shift of the top level at A, model vs GPAW's full SOC: 322 vs 321 meV ($\hat{n} \parallel c$), 75 vs 60 meV ($\hat{n}$ in plane)



Anomalous Hall conductivity for three Néel vectors

```
ahc = wb.calculators.static.AHC(Efermi=Efermi,
    degen_thresh=1e-4, smoother=smoother)
directions = {"x": (90, 0), "y": (90, 90),
              "z": (0, 0)} # theta, phi
grids = {"x": (48, 48, 32), "y": (96, 96, 64),
         "z": (48, 48, 32)}
for name, (theta, phi) in directions.items():
    system.set_soc_axis(theta=theta, phi=phi,
                        units="degrees")
    result = wb.run(system,
                    grid=wb.Grid(system, NK=grids[name]),
                    calculators={"ahc": ahc})
```

- $\hat{n} \parallel x$ and $\hat{n} \parallel c$: zero, as symmetry says
- $\hat{n} \parallel y$: up to 84 S/cm, sign change 0.48 eV below the valence band top E_v



Symmetry gives $\sin 3\varphi$

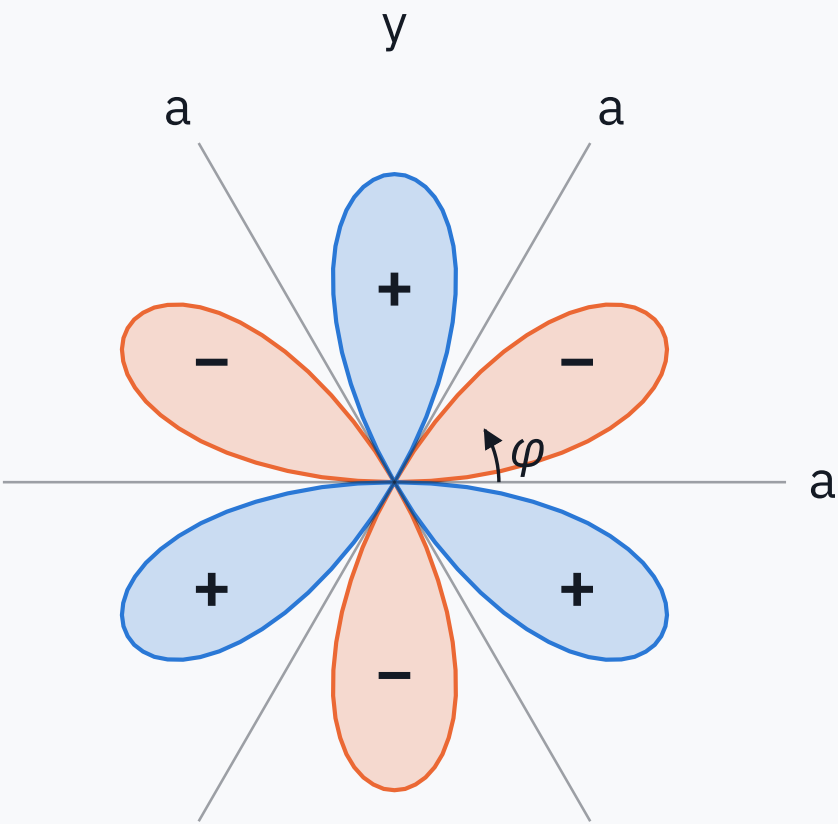
In-plane Néel vector at angle φ from a:

Operation	Swaps Mn	$\varphi \rightarrow$	So
turn 180° about a	no	$-\varphi$	$\sigma(-\varphi) = -\sigma(\varphi)$
turn 60° about c, shift c/2, then $\mathcal{T}$	yes	$\varphi + 60^\circ$	$\sigma(\varphi + 60^\circ) = -\sigma(\varphi)$

$$\sigma_{xy}(\varphi) = A_3 \sin 3\varphi + A_9 \sin 9\varphi + A_{15} \sin 15\varphi + \dots$$

A_m : amplitudes in S/cm

- Fixed by symmetry: zeros along every a axis, sign change every 60°
- Not fixed: the A_m ; $\sin 3\varphi$ need not dominate



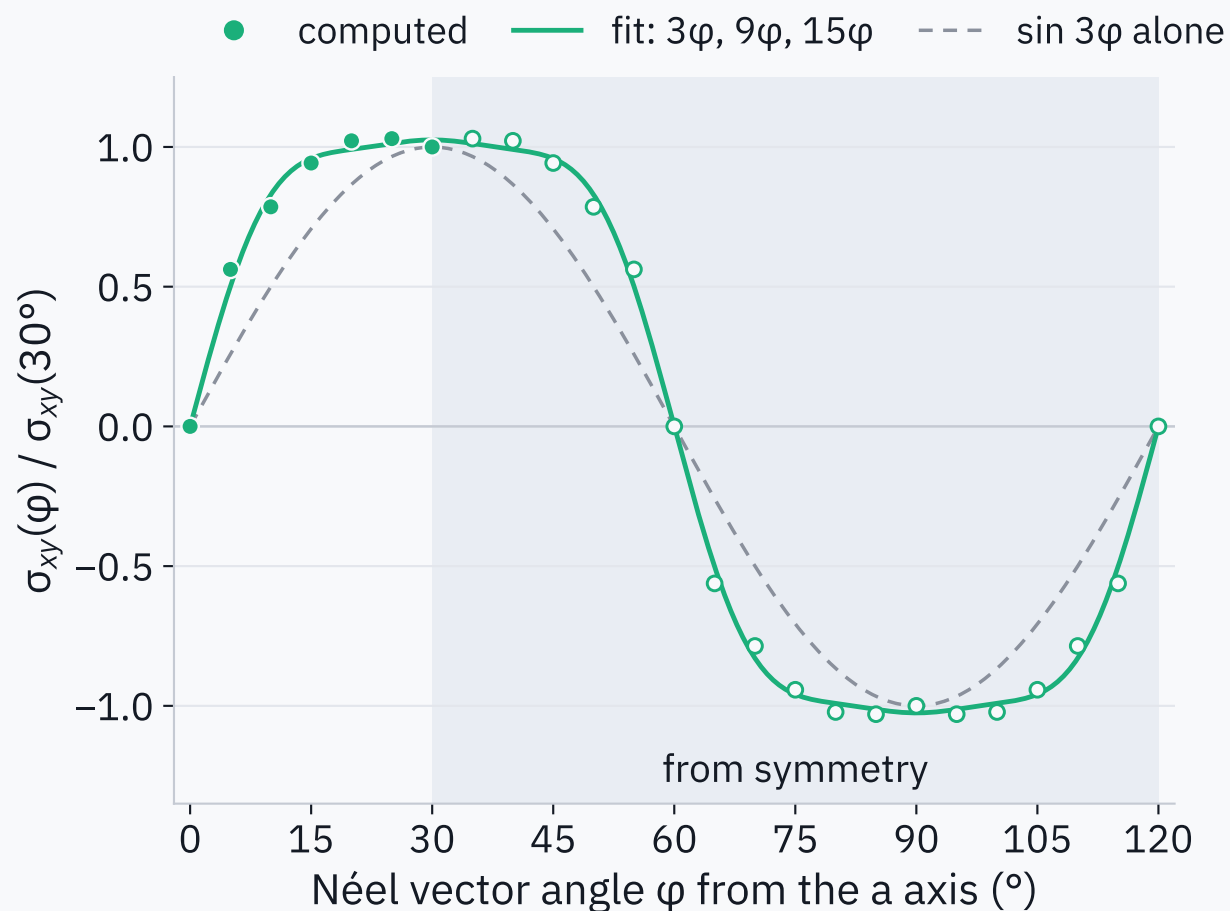
Sign of σ_{xy} against the direction of $\hat{n}$, $\sin 3\varphi$ only (sketch)

Rotating the Néel vector: exact zeros, but not a $\sin 3\varphi$ curve

One `set_soc_axis` per angle, no new DFT: 0–30° in 5° steps, 1 min

- $\sigma(-\varphi) = -\sigma(\varphi)$ and $\sigma(\varphi + 60^\circ) = -\sigma(\varphi)$ hold to $3 \cdot 10^{-7}$ S/cm
- $\sigma(5^\circ)/\sigma(30^\circ) = 0.56$, $\sin 3\varphi$ alone gives 0.26
- Fit: $A_3 = 172$, $A_9 = 31$, $A_{15} = 7$ S/cm

$E_F = E_V - 0.7$ eV ($6.6 \cdot 10^{21}$ holes/cm³), 48×48×32 grid



Filled: computed. Open: from the two symmetry relations

Your material next

```
calc = GPAW(...) # SCF
kpts = SpaceGroup.from_gpaw(calc)\
    .get_irreducible_kpoints_grid(grid)
calc.fixed_density(kpts=kpts, nbands=...) # NSCF
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=...)
wandata.wannierise(froz_min=..., froz_max=...,
    sitesym=True)
system = wb.SystemSOC.from_wannierdata(
    wandata, berry=True)
system.set_soc_axis(theta, phi)
wb.run(system, grid=..., calculators=...)
```

- Projections: the orbitals that make the bands you care about
- Frozen window: every state you need kept exactly
- k-grids: multiples of the DFT grid
- `degen_thresh` below the smallest splitting
- Check a few levels against GPAW first

Take-aways, and how to run it

- Symmetry: DFT on 7 of 36 and 24 of 144 k-points
- SOC on demand: any spin axis and strength, no new DFT
- Spin Hall plateau $e/2\pi$ at the real gap
- Altermagnets: Wannierise one spin channel, get both
- The Néel vector is just a parameter: symmetry gives the zeros, the bands give the size

RUN IT

github.com/yaroslav-zhumagulov/wannier-berri-demo

```
conda env create -f setup/environment.yml
conda activate wbdemo
bash setup/install_wannierberri.sh
python setup/check_setup.py
bash run_all.sh      # 6 min, or the notebooks
```

Apple Silicon: create the environment for osx-64 (README)

BACKUP

Details

Kane–Mele form · nodal planes of MnTe · uniform grids · timings ·
references

The Wannier SOC matrix has the Kane–Mele form

$$H_{\text{SO}} = i\lambda \sum_{\langle\langle ij \rangle\rangle} \nu_{ij} c_i^\dagger s_z c_j, \quad \Delta_{\text{SO}} = 6\sqrt{3} \lambda$$

- Second neighbours: $\lambda = 3.44 \text{ }\mu\text{eV}$, purely imaginary, sign ± 1 as in Kane–Mele
- $6\sqrt{3} \lambda = 35.7 \text{ }\mu\text{eV}$, full model $35.8 \text{ }\mu\text{eV}$
- $\Delta V^{x,y} = 0$: the mirror plane forbids Rashba terms, s_z is conserved
- No SOC between first neighbours: each bond lies in a vertical mirror

Shell	$ d $ (Å)	$ t $ (eV)	$ \Delta V^{x,y} $ (eV)	$ \Delta V^z $ (μeV)
1st, A–B	1.42	2.90	0	0
2nd, A–A	2.46	0.25	10^{-25}	3.44
3rd, A–B	2.84	0.26	10^{-26}	0
4th, A–B	3.76	0.02	10^{-26}	0
5th, A–A	4.26	0.04	10^{-25}	0
6th, A–A	4.92	0.03	10^{-25}	0.26

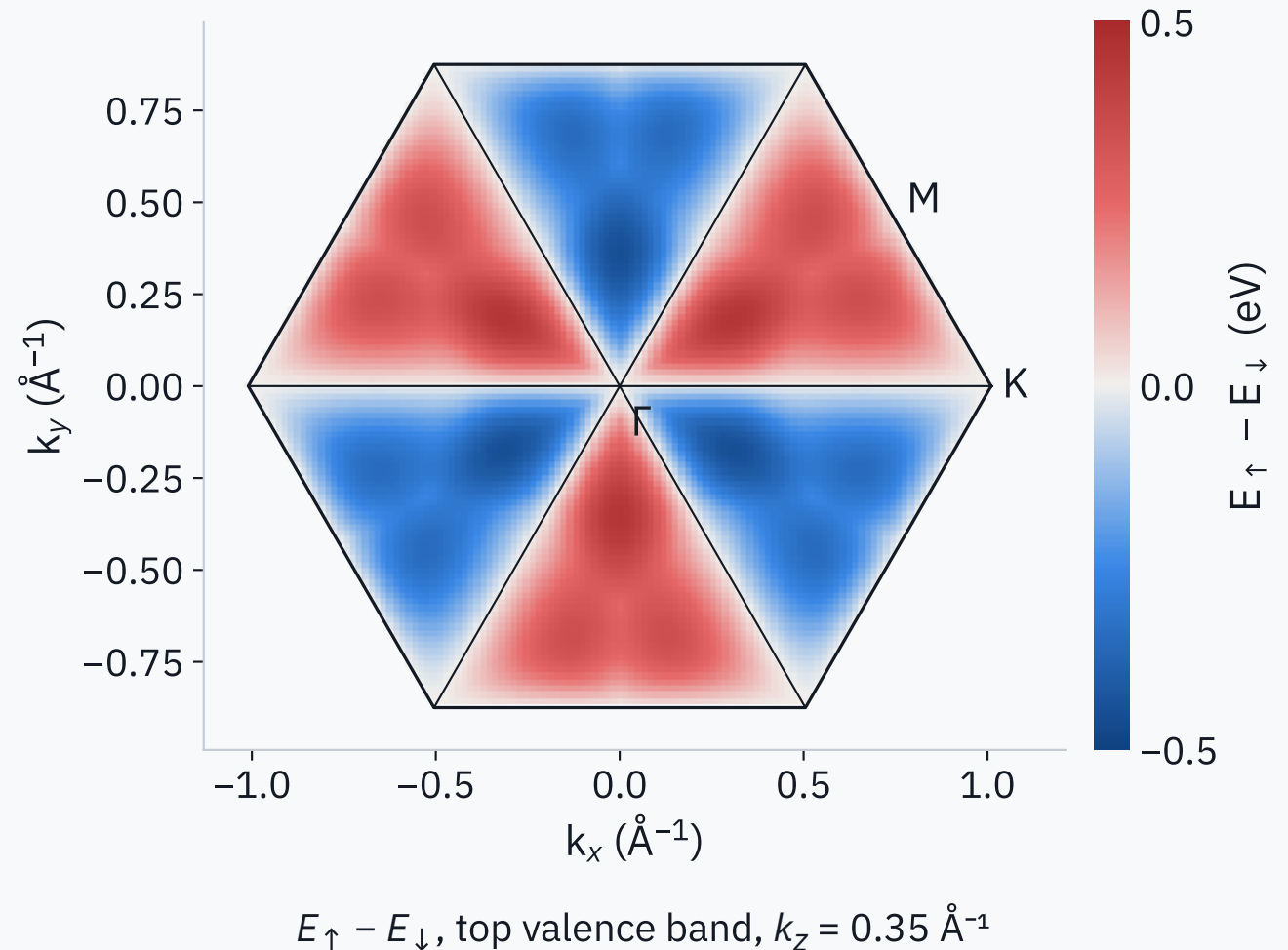
Largest element per shell of $H(\mathbf{R})$ and $\Delta V(\mathbf{R})$; nothing fitted

Four nodal planes make MnTe g-wave

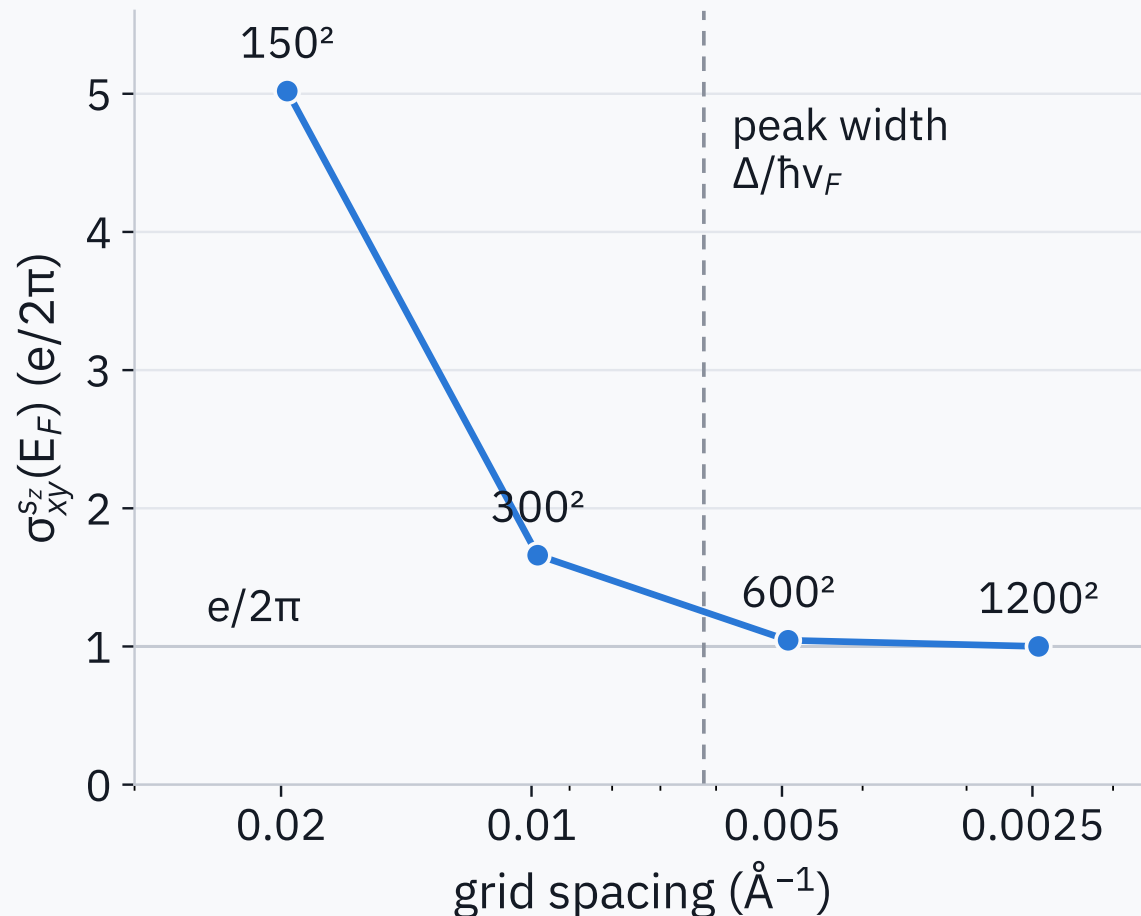
$E_{\uparrow}(\mathbf{k}) = E_{\downarrow}(R_g \mathbf{k})$ for a spin-flipping g , so the splitting vanishes on planes that R_g leaves in place:

- m_z : $k_z = 0$ and $k_z = \pi/c$
- three c -glides: three vertical planes through Γ –K

In the model: $|E_{\uparrow} - E_{\downarrow}| \leq 1.5 \cdot 10^{-14}$ eV on these planes, up to 0.95 eV along Γ –M at $k_z = 0.35 \text{ \AA}^{-1}$



Uniform grids overshoot until they resolve the peak at K



$\alpha = 1000$ (gap 36 meV), plain k-sum at E_F , 4 s for all four grids

- Spin Berry curvature: a peak at K, width $\Delta/\hbar v_F \approx 0.006 \text{ \AA}^{-1}$
- K is on every grid divisible by 6, so coarse grids give the peak too much weight
- 1200^2 : 1.0003 $e/2\pi$
- Outside the gap, tetrahedra remove the steps of a plain sum

Where the time goes

Step	Graphene	MnTe
GPAW SCF + NSCF, 4 MPI ranks	10 s	45 s
<code>WannierDataSOC.from_gpaw</code>	3.3 s	9.7 s
<code>wannierise</code>	2.1 s	9.7 s
<code>SystemSOC.from_wannierdata</code>	1.0 s	6.6 s
Conductivities (step 5)	28 s	63 s
Néel scan (step 6)	–	66 s
Animation frames	1 s	22 s
Whole notebook	1.1 min	4.2 min

Apple M2, 8 cores, 8 Ray workers:
6 minutes for everything, try-its
and bonus steps included

With the Wannier model in hand, a
new spin axis, SOC strength or
Fermi level costs seconds

References

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